Business Requirements Document (BRD): Gondal Fulbe Integrated ERP System

Table of Contents

1. Background & Objectives
2. User Roles and Permissions Matrix
3. Functional Modules Specifications
4. Integration Specifications (Sebore Platform)
5. Non-Functional Requirements

Glossary

Term	Definition
MCC	Milk Collection Centre
BVN	Bank Verification Number (Nigeria)
CEP	Community Engagement Program
EPO	Executive President's Office
ERP	Enterprise Resource Planning
JWT	JSON Web Token
NDPR	Nigeria Data Protection Regulation
OSS	One-Stop Shop
RBAC	Role-Based Access Control
TLS	Transport Layer Security
USSD	Unstructured Supplementary Service Data

1.0 Background & Objectives

Gondal Fulbe is a community-driven dairy cooperative with current operations that includes 5 Milk Collection Centers (MCCs): Mayo, Yola, Jabbi Lamba, Mubi, and Sunkani. The cooperative operates on four strategic pillars: Production Excellence, Infrastructure Expansion, Institutional Strengthening, and Innovation. The following are key objectives for this ERP.

- Digitize end-to-end milk value chain tracking
- Enable automated payments and reconciliation with Sebore
- Provide real-time visibility to leadership (Board, Executive Director)
- Ensure scalability for 2030 targets

2.0 User Roles and Permissions Matrix

Role	Description	Modules Access	Permissions (CRUD)
Farmer/Vendor	Milk supplier, cooperative member	Farmer Portal (USSD/Web/App)	Read: own milk history, payment receipts
Community Cooperative Leader	Elected community representative	Community Dashboard	Read: community aggregates; Update: member verification
Field Delivery Lead	Gondal field staff at community level	Milk Collection, Logistics, Requisition, Farmer Registration	Create/Read/Update: assigned communities
Center Manager	Manages specific MCC	Center Ops, Costs, Inventory, Requisition, Milk Collection	Create/Read/Update: own center; Approve: center costs
Component Lead	EPO-level pillar manager	Pillar-specific data, Requisition Approval	Read: pillar reports; Approve: requisitions (tier 1)
Extension Agent	Field trainer	Extension Management	Create/Read: visit logs, training events
Finance Officer	Payments and reconciliation	Payments, Reconciliation, Center Costs, Requisition (financial)	Create/Read/Update: payments; Approve: costs (tier 2)
Executive Director	Top executive	ALL modules, Strategic Dashboard	Full access; Final approval (tier 3)
Board Member	Farmer-elected/expert oversight	Dashboard, Reports	Read-only: strategic KPIs, financial summaries
System Administrator	Technical administration	User Management, Audit Logs, System Config	Full system configuration

Permission Key: C=Create, R=Read, U=Update, D=Delete (limited)

4. Functional Modules Specifications

Module 1: Farmer & Cooperative Management	**Purpose:** Central registry of all farmers and cooperatives with demographic tracking. **Data Fields:** • Farmer: Unique ID (auto), full name, gender, date of birth, phone, photo, cooperative ID, bank name, account number, BVN (optional), GPS coordinates, registration date, status (active/inactive), digital payment enabled (Y/N) • Cooperative: ID, name, location, leader name/phone, formation date, member count, average daily supply **Features:** • Bulk import/export (Excel/CSV) • Duplicate detection • Search/filter by: name, ID, cooperative, gender, status, date range • Document upload (registration forms, ID copies) **Business Rules:** • Each farmer belongs to exactly one cooperative • Farmers marked inactive after 60 days no supply (with notification) • Digital payment flag enables auto-payouts
Module 2: Milk Collection & Quality	**Purpose:** Record daily milk intake with quality grading. **Data Fields:** • Collection ID (auto), Date & time (default now), MCC (Mayo, Yola, Jabbi Lamba, Mubi, Sunkani) + expandable, Farmer ID (search/select), • Quantity (litres, 2 decimals) • Quality Grade: A (Premium), B (Standard), C (Rejected) • Temperature (optional, °C) • Rejection reason (if applicable) • Recorded by (user ID- auto) • Collection batch ID (for grouping) **Features:** • Quick entry: default to last 10 farmers for speed • Daily summary per MCC (total litres, avg quality, farmer count) • Offline capability: mobile app syncs when online • Photo attachment (milk quality evidence) • Reporting filters: date range, MCC, community, farmer etc. ***Integration:*** *Data pushed to Sebore daily (see Section 5)* **Business Rules:** • Quality Grade A: >4% fat, <20°C • Quality Grade B: 3-4% fat, 20-25°C • Grade C: auto-rejection • Farmer notified of rejection via SMS

Module 3: Logistics & Transport	**Purpose:** Manage milk transportation from MCCs to Sebore processor. **Data Fields:** • Trip ID (auto), Date, MCC source, Destination (default: Sebore plant), Rider ID (from rider registry), Vehicle registration, Departure time, Arrival time, Litres transported (auto-calc from collections), Fuel cost (NGN), Other expenses (NGN, with description), Status: Scheduled, In Transit, Completed, Delayed, Delivery note attachment (photo of waybill) **Rider Registry:** • Rider ID, name, phone, license number, vehicle type, vehicle reg, status **Features:** • Trip scheduling view (calendar) • Cost per litre calculation • Rider performance reports (on-time %, cost efficiency) • Expense tracking • Reporting filters: date range, rider, MCC, status **Business Rules:** • Trip cannot be completed without matching collection total • Fuel cost auto-calculates based on distance (optional)
Module 4: Center Operations & Cost Management	**Purpose:** Track operational expenses per MCC. **Cost Categories:** • Labour (casual workers), Cleaning supplies, Maintenance & repairs, Utilities (electricity, water), Rent, Miscellaneous (with description) **Data Fields:** • Cost entry ID, MCC, Category, Amount (NGN), Description/notes • Receipt attachment, Status: Draft, Submitted, Approved, Rejected, Paid, Approved by (user, date), Entered by **Features:** • Weekly cost summary report per center • Month-over-month comparison • Budget vs actual tracking (if budgets configured) • Approval workflow: 1. Center Manager submits 2. Component Lead reviews 3. Finance Officer/ED approves • Notification at each step **Reporting:** • Total operational cost per litre • Cost breakdown by category • Center ranking by cost efficiency

Module 5: Requisition & Approval Workflow	**Purpose:** Request and approve supplies/services for centers. **Requisition Fields:** • Requisition ID (auto) • Request date • Requester (user) • Center • Title • Description • Items table: item name, quantity, unit, estimated cost, purpose • Total estimated cost • Priority: Low, Medium, High, Emergency • Attachments (up to 5 files) • Status: Pending, Supervisor Approved, Manager Approved, Rejected, Paid, Completed • Approval chain log (user, date, action, comments) • Payment reference (when paid) **Approval Workflow:** text Step 1: Requester submits → Status: Pending Step 2: Immediate Supervisor reviews → Approve/Reject/Request Changes If Approved → Status: Approved Step 3: Component Lead Approves, if approved → Status Approved Step 4: Finance/Manager reviews (depending on amount) • < ₦50,000 → Finance Officer approves • ₦50,000- ₦200,000 → Component Lead approves • > ₦200,000 → Executive Director approves Step 5: Approved → Finance Officer marks Paid Step 6: Requester confirms receipt → Status: Completed **Features:** • Dashboard: My Requisitions, Pending Approvals • Email/in-app notifications • Audit trail • Rejection reason required • SLA tracking (time to approve)
Module 6: Payment & Reconciliation	**Purpose:** Process all payments and reconcile with Sebore. **Payment Types:** 1. Farmer Milk Payments (weekly) 2. Logistics/Rider Payments (weekly) 3. Center Operational Costs (as approved) 4. Requisition Payments (as approved) **Farmer Payment Flow:** 1. System matches milk collection records with payout data 2. Discrepancies flagged for review

	3. Finance Officer reviews and approves payment batch 4. Batch executed via payment gateway (Paystack/Monipoint/Monify) 5. Transaction IDs recorded 6. Farmers notified via SMS 7. Payment receipts generated **Payment Batch Fields:** • Batch ID, Batch type, Period start/end, Total amount, Number of transactions, Status: (*Draft, Approved, Processing, Completed, Failed*), Approved by, Executed date. **Individual Payment Receipt:** • Farmer name/ID, Period, Gross amount, Deductions (credit repayments), Net amount, Payment date, Transaction ID, Downloadable PDF **Features:** • Manual payment entry (for exceptions) • Bulk payment file generation (bank format) • Payment retry for failures • Reconciliation reports • Audit log of all payment actions **Business Rules:** • Farmers with digital payment flag = auto-pay from ERP • Others require manual bank transfer (receipt upload)
Module 7: One-Stop Shop (Inventory & Sales)	**Purpose:** Manage inputs inventory and sales to farmers. **Product Catalog:** • Product ID, Name, Category (Feed, Veterinary, Equipment etc.), Unit (kg, litre, piece), Price (NGN), Reorder level, Description, Supplier. **Inventory Transactions:** • Transaction ID, Type: (Stock In, Stock Out, Adjustment), Product • Quantity, Date, Center, Reference (purchase order, sales order), Unit cost (for stock in), Batch/expiry (optional) **Sales:** • Sales ID, Date, Farmer ID, Products (list with quantity, price), Total amount, Payment method: Cash, Credit, Mobile Money, Credit flag (if farmer takes on credit) **Features:** • Low stock alerts (email to Center Manager) • Stock valuation • Daily/Weekly/Monthly sales summary • Stock reconciliation (physical count vs system), additional note/reasons for mismatch. • Reporting: fast-moving items, slow-moving items. • Credit tracking

Module 8: Extension Management	**Purpose:** Track extension agent activities and farmer training. **Agent Profile:** • Agent ID, Name, Phone, Assigned communities/centers, Join date, Supervisor **Field Visit Log:** • Visit ID, Agent, Date, Farmers visited (list), Topics covered (multi-select: Animal Health, Feeding, Hygiene, Breeding, Record Keeping), Notes, Photos (up to 3) • Follow-up required (Y/N), Follow-up date **Training Event:** • Event ID, Title, Date, Location, Center, Facilitator(s), Topics, Farmers attended (list), Materials distributed (list with quantity), Photos. **Reports:** • Visits per agent (weekly/monthly) • Topics coverage • Farmer participation • Impact on productivity (future: link to milk data) **Business Rules:** • Minimum 2 visits per agent per week target • Follow-up tasks auto-create reminder

Module 9: Reporting & Dashboard	**Purpose:** Role-based analytics and operational reports.
	Executive Dashboard (Executive Director, Board):
	• KPIs cards:
	  o Active farmers
	  o Daily milk collection
	  o Financial inclusion rate
	  o Centers operational
	• Charts:
	  o Milk collection trend (daily, weekly, monthly)
	  o Payments processed (weekly total)
	  o Top 5 performing centers (by volume)
	  o Gender distribution pie chart
	• Alerts: centers below target, payment failures
	Operational Reports (Component Leads, Managers):
	1. Milk Collection Report
	  o Filters: date range, center, community, farmer
	  o Output: farmer, quantity, quality, total
	  o Export: PDF, Excel
	2. Logistics Summary
	  o Filters: date range, rider, center
	  o Output: trips, litres, cost/trip, cost/litre
	3. Center Operations Cost
	  o Filters: center, week/month

	o Output: cost categories, total, comparison 4. Requisition Status o Filters: center, status, date o Output: requisitions, approval time, total cost 5. Payment History o Filters: date, type, farmer/rider o Output: payments, status, transaction IDs 6. Extension Activity o Filters: agent, date, center o Output: visits, farmers reached, topics 7. Inventory Report o Stock levels, sales, reorder alerts Report Scheduling: • Daily email to managers (summary) • Weekly PDF to Board and ED • Custom scheduling

5. Non-Functional Requirements

Category	Requirement
Security	Role-based access control (RBAC); TLS 1.3 for all transmissions; AES-256 encryption at rest; API keys with rotation policy; audit logs for all financial transactions; 2FA for finance/admin users
Scalability	Support 60,000 farmers, 35 centers, 200+ concurrent users; database sharding ready; cloud auto-scaling.
Availability	99.5% uptime (core hours 6am-8pm); planned maintenance window Sunday 2am-4am. With prior notification.
Performance	Page load < 3 seconds; report generation < 10 seconds; API response < 2 seconds; batch processing < 5 minutes for 10,000 records
Offline Capability	Mobile app for field staff: offline data entry, sync when online; conflict resolution
Mobile Access	USSD/MobileApp for farmers: check balance, last payment; responsive web for all users
Backup & Recovery	Automated daily backups; 30-day retention; point-in-time recovery; disaster recovery plan
Audit Trail	All create/update/delete on financial records, farmer data, user accounts logged with timestamp, IP, user
Compliance	Nigeria Data Protection Regulation (NDPR); GDPR principles for data minimization
Integration	API-first design; API documentation; webhook support